

VF6

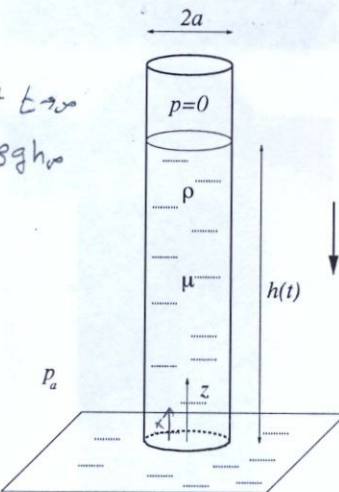
The bottom end of an empty vertical tube of radius a , closed at the top, is put in contact with a pool of oil. Because of the ambient overpressure, the liquid begins to flow into the tube, forming a column of increasing height $h(t)$ whose evolution in time is to be investigated assuming that the motion is dominated by viscosity. In particular:

- Obtain the value of the height h_∞ corresponding to the equilibrium position, reached asymptotically for large times.
- Give the condition that determines whether the motion is dominated by viscosity.
- Obtain the evolution of $h(t)$ as well as the pressure distribution along the pipe $p(x, t)$ for $0 < x < h$.
- Compute the force acting on the pipe as a function of time $\vec{F} = F_z(t)\vec{e}_z$.

1) IN THE LIQUID IN EQUILIBRIUM FOR $t \rightarrow \infty$

$P + \rho g z = \text{CONSTANT} = P_a = \rho g h_\infty$

$h_\infty = \frac{P_a}{\rho g}$



2) IF DOMINATED BY VISCOSITY, $V_L \sim \frac{D^2}{L} \frac{P_a}{\mu}$

$\frac{O(\rho \frac{dV_L}{dt})}{O(\mu \nabla^2 V)} \sim \frac{D^2/(M/s)}{t_0} \sim \frac{D^2/(M/s)}{L/V_L} \sim \frac{\rho V_L D D}{\mu L}$

$\frac{O(\rho \vec{V}_L \cdot \nabla \vec{V})}{O(\mu \nabla^2 \vec{V})} \sim \frac{\rho V_L D D}{\mu L}$

CRITERION $\frac{\rho V_L D D}{\mu L} \sim \left(\frac{\rho D^4 P_a}{\mu^2 h_\infty^2} \sim \frac{\rho^3 g^2 D^4}{\mu^2 P_a} \ll 1 \right)$

3) $0 = -\frac{\partial P}{\partial x} + \frac{\mu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V_x}{\partial r} \right)$ $\int_{r=0}^{r=a}, V_x=0$

$V_x = \frac{P_x}{4\mu} (a^2 - r^2) \rightarrow Q = \int_0^a 2\pi r V_x dr = \frac{\pi}{8\mu} P_x a^4$

$P_x(t) = -\frac{\partial P}{\partial x} = \frac{8\mu Q(t)}{\pi a^4} \Rightarrow P = P_a + \rho g z = -\int \frac{8\mu Q}{\pi a^4} dx + C_1 =$

$C_1 - \frac{8\mu Q}{\pi a^4} x$

$x=0, P=P_a \rightarrow C_1=P_a$

$P + \rho g z = P_a - \frac{8\mu Q}{\pi a^4} z$

$x=h, P=\rho g h$

$P_a - \rho g h = \frac{8\mu Q}{\pi a^4} h$

$Q = \pi a^2 \frac{dh}{dt} = \frac{P_a - \rho g h}{h} \frac{\pi a^4}{8\mu}$

$\frac{dh}{dt} = \frac{P_a}{8\mu} \frac{1 - h/h_\infty}{h}$

$\int_0^h \frac{h dh}{1 - h/h_\infty} = \int_0^t \left(\frac{P_a}{8\mu} \right) dt$

$\frac{P_a}{8\mu h_\infty} t = \ln\left(\frac{1}{1 - h/h_\infty}\right) - \frac{h}{h_\infty}$

$h(t) \mid \text{As } t \rightarrow \infty, h \rightarrow h_\infty$

4)



$\tau_p = -\mu \frac{\partial V_x}{\partial r} = \frac{P_x a}{2} = \frac{a}{2} \frac{P_a - \rho g h}{h}$

$F_p = 2\pi a h \frac{a}{2} \frac{P_a - \rho g h}{h} = \pi a^2 (P_a - \rho g h)$

THE TOTAL FORCE IS

$\pi a^2 (P_a - \rho g h) \vec{e}_z - P_a \pi a^2 = -\rho g h \pi a^2 \vec{e}_z$

VISCOS FENCE ON THE WALL

PRESSURE FENCE ON THE TOP END